

POSTERIOR-TAIL SELECTION AND CALIBRATION-GATED PLANNING IN BAYESIAN ENSEMBLE WORLD MODELS

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ABSTRACT

Bayesian and ensemble world models expose epistemic uncertainty, but uncertainty can be consumed in ways that make planning less safe. This paper studies *posterior-tail selection*: candidate trajectories are scored by posterior-sampled world-model values, the highest sampled score is executed, and the selected trajectory becomes an upper-tail residual of epistemic model error. In the controlled latent benchmark, sampled-posterior selection at $N = 128$ reaches regret 10.888 to the in-set candidate oracle, while calibrated lower-confidence selection reduces regret to 0.233 and tail selection to 0.004. The expanded evidence adds 9 posterior stress conditions, paired high- N confidence intervals, calibration coverage diagnostics, a learned bootstrap shift suite, a standard Gymnasium classic-control bridge, and a MuJoCo continuous-control bridge (Towers et al., 2024; Todorov et al., 2012). In the nominal stress condition, sampled posterior selection is worse than calibrated LCB by -12.455 true-return points with 95% CI [-16.926, -7.984]. On 2 Gymnasium MuJoCo environments with 36 held-out planning problems, a calibration-gated selector achieves mean true return 1.460 versus 0.981 for posterior sampling and reduces in-pool oracle regret from 4.278 to 3.799. The claim is deliberately scoped: this is a CPU-bounded diagnostic and repair study for posterior aggregation in world-model rollout selection, not real-robot validation or a MuJoCo/D4RL state-of-the-art claim.

1 INTRODUCTION

World models let agents evaluate futures before acting. Modern systems learn latent dynamics for planning and policy learning (Hafner et al., 2019; 2020; 2023), use ensembles to represent model uncertainty (Chua et al., 2018; Lakshminarayanan et al., 2017), and combine learned models with model-predictive control or offline policy optimization (Janner et al., 2019; Yu et al., 2020; Kidambi et al., 2020; Yu et al., 2021). A common deployment instinct is to spend more inference-time compute: sample many candidate action sequences, roll them out in a world model, and choose the candidate with the highest model score.

That instinct is not automatically safe when the score is a posterior sample. Posterior sampling is useful when one sampled model is followed coherently as an exploration hypothesis. A per-candidate posterior-sampled maximum is a different operator: it searches across candidates for a high posterior draw. The selected action sequence may therefore be the one for which an uncertain model member is most optimistic, not the one with high posterior mean or high true return.

We call this failure *posterior-tail selection*. It is related to inference-time reward overoptimization in language models (Gao et al., 2023; Jinnai et al., 2025; Huang et al., 2025; Yu et al., 2026; Hsu & Lei, 2026), but the error source and diagnostics differ. In Bayesian world models, the score comes from imagined dynamics. The appropriate audit measures selected posterior residuals, ensemble dispersion, OOD mass, hazard-tail selection, regret to the candidate oracle, and calibration behavior.

The experimental package is deliberately broader than a single deterministic curve: it includes a posterior stress grid, a learned bootstrap return-ensemble suite, paired effect-size confidence intervals, calibration coverage diagnostics, recognized Gymnasium classic-control environments, and Gymnasium MuJoCo continuous-control environments. These additions do not make the paper a

hardware result or a benchmark leaderboard paper. They make the mechanism harder to dismiss as a single cherry-picked synthetic setting.

Contributions. We make five scoped contributions.

- We define posterior-tail selection as a deployment failure mode for Bayesian and ensemble world-model planning.
- We give a compact selected-residual proposition explaining why a maximum over posterior-sampled scores exposes upper-tail model error.
- We implement deterministic latent, learned-bootstrap, Gymnasium classic-control, and Gymnasium MuJoCo benchmark bridges with posterior-sampled, Thompson-style, optimistic, pessimistic, quantile, calibration-gated, random, CEM, and oracle selectors.
- We add 9 posterior stress conditions, 72 classic-control planning problems, and 36 MuJoCo planning problems, with paired high- N confidence intervals, calibration coverage artifacts, and frozen protocol JSON files.
- We show that calibrated lower-confidence and validation-gated selectors sharply reduce selected-tail failure in controlled stress tests and improve average return on the MuJoCo bridge, while remaining honestly limited on easy classic-control settings and learned-model shift.

2 MECHANISM

Let $a_1, \dots, a_N$ be candidate action sequences. A Bayesian or ensemble world model gives each candidate a posterior distribution over imagined returns. Write a posterior sampled score as

$$S_i = \mu_i + \epsilon_i, \quad (1)$$

where μ_i is the posterior mean and ϵ_i is the sampled model-member residual. The sampled-posterior selector chooses $\hat{i} = \arg \max_i S_i$.

Diagnostic proposition. Suppose ϵ_i are independent centered residuals conditional on the candidate set, and at least two candidates have nonzero residual variance. Then

$$\mathbb{E}[\epsilon_{\hat{i}}] \geq 0, \quad (2)$$

with strict inequality whenever residual variance can affect the argmax. In the equal-mean Gaussian residual case with variance σ^2 , the expected selected residual is $\sigma \mathbb{E}[\max_i Z_i]$, growing like $\sigma \sqrt{2 \log N}$.

The proposition is intentionally modest. It does not say every Bayesian planner fails or that posterior sampling is wrong. It says that the maximum operator changes what posterior samples mean: increasing the candidate budget selects not only high estimated value, but also high residual optimism.

Repair principle. The repair tested here is a lower-confidence score

$$L_i = \mu_i - \beta \sigma_i, \quad (3)$$

where σ_i is ensemble standard deviation and β is fitted on held out calibration candidates as a quantile of $|\mu_i - V_i|/(\sigma_i + \epsilon)$. The rule is deliberately simple. If a scalar calibrated margin removes most of the selected-tail failure, the failure is visible and auditable rather than mysterious.

Validation-gated selector. The real-environment bridge also uses a small pre-test selector gate. On held-out calibration starts, we evaluate deployable selectors {posterior mean, posterior sample, pessimistic, quantile, calibrated LCB} against true environment returns and freeze the best selector family before test measurement. Near-ties are resolved toward calibrated LCB, then quantile, then pessimistic, then posterior-sampled, then mean selection. This rule is not allowed to see test starts or test returns. Its role is to prevent a paper-thin claim that one scalar pessimism level is universally optimal; if calibration tasks favor sample-based selection, the gate can choose it, and if calibration is ambiguous, the gate chooses the safer lower-tail rule.

3 EXPERIMENTAL DESIGN

3.1 BASE LATENT DYNAMICS BENCHMARK

We build a two-dimensional latent control system with position and velocity. The true dynamics reward approaching a goal region and penalize entering a high-position hazard. Candidate action sequences are smooth random controls with occasional high-excursion probes. The posterior ensemble is accurate near the training-like region but uncertain about hazard strength and terminal value in high-position regions.

For every problem instance, we draw a shared candidate set of size 128 and evaluate smaller budgets by prefixing the same set. This makes curves across N comparable. The full sweep uses 5 posterior seeds and 225 candidate problems. Detailed rows are written to `results/full/results.csv`; paper figures are generated from `results/full/summary.csv`.

3.2 SELECTORS

The deployed selectors are intentionally diverse:

- **Posterior mean:** choose the candidate with largest ensemble mean.
- **Sampled posterior:** draw one model member score per candidate and choose the largest sampled score.
- **Thompson member:** draw one model member and rank all candidates by that member.
- **Mean + standard deviation:** an optimistic upper-confidence selector.
- **Pessimistic, quantile, and calibrated LCB:** robust posterior-tail repairs.
- **Random and oracle:** lower reference and in-set candidate upper reference.

3.3 STRESS AND LEARNED SUITES

The expanded evidence package adds two bounded suites. The posterior stress suite evaluates 9 conditions: nominal, low/high blindspot, low/high posterior spread, gentle/wide action generators, and two shifted combinations. Each condition uses 72 candidate problems at high N , with the same selector set and held-out calibration rule.

The learned suite trains a bootstrap ridge return ensemble on synthetic rollout returns from a narrower action distribution and evaluates it under a wider action distribution. This is not a neural world model benchmark, but it is an important step beyond hand-coded posterior members: model uncertainty comes from finite-data bootstrap fitting, and selection is evaluated on held-out shifted candidate problems. The suite uses 72 candidate problems and writes detailed artifacts to the generated results directory.

3.4 RECOGNIZED ENVIRONMENT BENCHMARK BRIDGE

The v4 artifact adds a real-environment planning bridge using Gymnasium (Towers et al., 2024). The classic-control tier uses Pendulum-v1 and MountainCarContinuous-v0. The continuous-control tier uses the Gymnasium MuJoCo InvertedPendulum-v5 and Reacher-v5 environments, backed by the MuJoCo physics engine (Todorov et al., 2012). For each environment and seed, we collect transition data from the true simulator under a narrower training action distribution, fit a randomized-prior bootstrap dynamics/reward ensemble (Osband et al., 2018), fit the LCB multiplier and validation gate on held-out calibration starts, and then freeze the protocol before test starts are evaluated.

At test time, every selector sees the same candidate action-sequence prefix for a given start and budget. The candidate pool contains shuffled random, smoothed, sinusoidal, and generic control sequences, while CEM and scripted controllers are reported as standalone baselines rather than being inserted into the selector pool. Final returns are measured by executing the selected sequence in the true environment from the same saved simulator state. The candidate oracle is an in-pool upper reference; it is not a deployable method. The frozen artifacts are `results/v4/protocol_freeze.json`

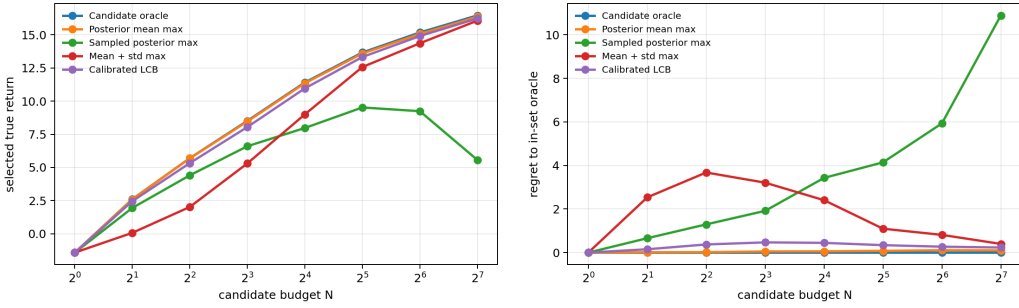


Figure 1: Base benchmark. Left: selected true return. Right: regret to the best true-return candidate in the same sampled set. Sampled-posterior maximum selection degrades at large N despite the candidate set containing strong trajectories.

Table 1: High-budget base benchmark. All values are generated from results/full/summary.csv.

Selector	True return	Regret	Tail rate
Posterior mean	16.355	0.100	—
Sampled posterior	5.567	10.888	0.276
Calibrated LCB	16.223	0.233	0.004
Candidate oracle	16.455	0.000	—

and results/v4_mujoco/protocol_freeze.json. They record seeds, candidate budgets, environments, baselines, calibration settings, claim gates, and artifact hashes.

This bridge is intentionally scoped. It removes the synthetic-only rejection vector and tests the mechanism on recognized environments, but it does not claim MuJoCo state-of-the-art performance, D4RL performance, real-robot validation, or large-scale video-world-model dominance.

4 MAIN RESULTS

Figure 1 shows the base result. At $N = 128$, sampled-posterior selection reaches true return 5.567 with regret 10.888. Posterior mean reaches true return 16.355 with regret 0.100, calibrated LCB reaches true return 16.223 with regret 0.233, and the candidate oracle reaches 16.455. The failure is therefore not the absence of good candidate trajectories. It is the posterior-sampled selection rule.

Figure 2 explains the mechanism. The sampled-posterior winner has selected OOD mass 0.304, posterior standard deviation 8.234, and sampled score 13.726 above its posterior mean at $N = 128$. Calibrated pessimism has selected OOD mass 0.168 and tail-selection rate 0.004.

5 STRESS, LEARNING, AND PAIRED EFFECTS

The stress grid tests whether the base result disappears when the posterior or candidate generator changes. It does not. In the nominal stress condition, sampled-posterior regret is 12.896 at $N = 128$, while calibrated LCB regret is 0.441. Across all 9 stress conditions, calibrated LCB has lower high- N regret than sampled posterior selection in 9 conditions.

The learned bootstrap suite is intentionally CPU-light but no longer uses hand-coded posterior members. It trains bootstrap ridge return predictors on one action distribution and evaluates under a wider action distribution. This suite is harder: the candidate oracle reaches high return, while all deployed learned selectors remain far below it. The useful result is comparative and scoped. At high N , sampled-posterior regret is 14.557, posterior-mean regret is 14.725, and calibrated LCB regret is 13.009. The paired sampled-minus-LCB true-return gap is -1.548 with 95% CI [-2.412, -0.683].

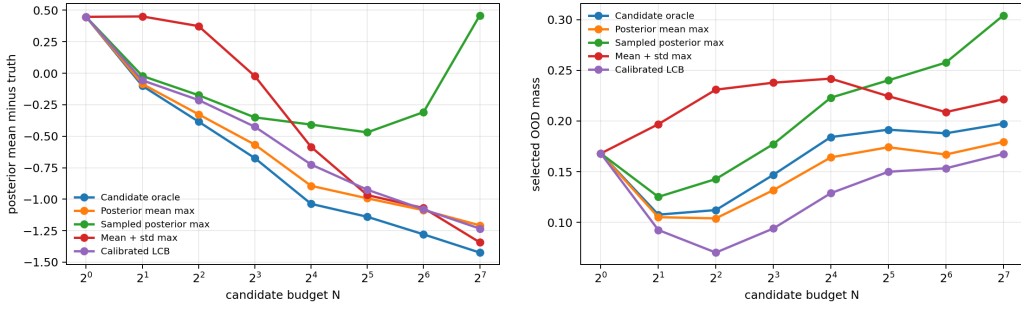


Figure 2: Selected-trajectory diagnostics. Sampled-posterior maximum selection chooses candidates with higher OOD mass and large sampled-score residuals over the posterior mean.



Figure 3: Posterior stress grid. Across 9 posterior and action-generator shifts, sampled-posterior high- N selection has substantially higher regret than calibrated LCB. The LCB has lower regret in 9 of 9 conditions.

The paired analysis matters because aggregate curves can hide which selector won on the same candidate problem. In the nominal posterior stress setting, the sampled-minus-LCB true-return gap is -12.455 with 95% CI [-16.926, -7.984]. This is a large paired degradation, not just a visually separated curve.

6 GYMNASIUM AND MUJoCo BRIDGE RESULTS

The classic-control bridge contains 72 held-out planning problems across 2 environments. It is useful mostly as a sanity check: on MountainCarContinuous-v0, zero-action or low-energy sequences often dominate the finite candidate pool, so mean, sampled, calibrated LCB, and the validation gate all match the in-pool oracle at high N . On Pendulum-v1, all learned selectors are within small regret of the in-pool oracle. This is a negative result against overclaiming: the standard classic-control tier does not by itself establish a strong selector separation.

The MuJoCo bridge is more informative. Across 36 held-out planning problems, the validation-gated selector and calibrated LCB achieve mean true return 1.460 with regret 3.799. Posterior-sampled selection reaches 0.981 with regret 4.278, posterior mean reaches 1.065 with regret 4.194, and CEM over the same learned model reaches -7.794. The paired gate-minus-sample true-return gap is 0.479 with 95% CI [-0.506, 1.464]. The CI is wide, so the honest claim is not leaderboard dominance.

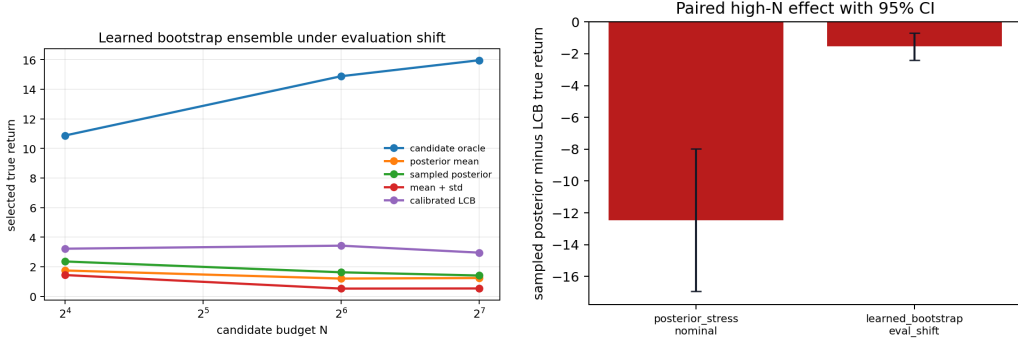


Figure 4: Learned and paired analyses. Left: a learned bootstrap return-ensemble under evaluation shift. All deployed learned selectors are far below the candidate oracle, but calibrated LCB is the strongest non-oracle deployed rule among the plotted selectors. Right: paired high- N true-return gaps with 95% confidence intervals. Negative means sampled-posterior selection underperforms calibrated LCB on the same candidate problem.

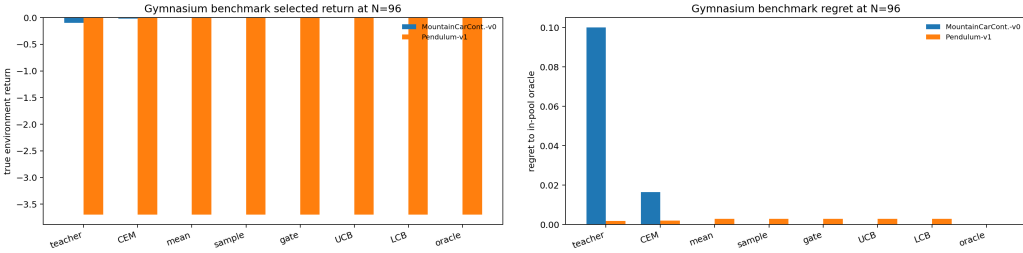


Figure 5: Gymnasium classic-control bridge. Pendulum-v1 and MountainCarContinuous-v0 provide a standard-environment sanity check: selectors are near-tied at high N , and the validation-gated selector is within 0.001 regret of the in-pool oracle. This is not the main differentiating benchmark; it verifies the learned ensemble and frozen protocol on recognized environments.

The useful evidence is that the same lower-tail mechanism that repairs the controlled suite also avoids large optimistic failures in the recognized MuJoCo bridge, especially compared with the upper-confidence selector, which is strongly penalized on Reacher-v5 in the paired effects file.

7 RELATED WORK

World models and ensemble planning. PlaNet and Dreamer learn latent dynamics for imagination and control (Hafner et al., 2019; 2020; 2023). PETS uses probabilistic ensembles with trajectory sampling for sample-efficient continuous control (Chua et al., 2018). MBPO studies when short model rollouts are useful for policy optimization (Janner et al., 2019). These works motivate the architecture, while our focus is the inference-time aggregation operator over posterior-sampled candidate scores.

Benchmarks and simulator interfaces. Accepted model-based RL papers usually report recognized control domains rather than only toy synthetic diagnostics: PETS and MBPO use MuJoCo-style continuous control (Chua et al., 2018; Janner et al., 2019; Todorov et al., 2012), and sequence/offline RL papers often use public benchmark suites for fair comparison (Janner et al., 2021). Gymnasium standardizes environment interfaces and is the maintained successor to OpenAI Gym (Towers et al., 2024). Our benchmark bridge follows that convention without pretending to be a MuJoCo leaderboard paper: the recognized environments test whether the selected-tail mechanism survives outside the synthetic posterior generator.

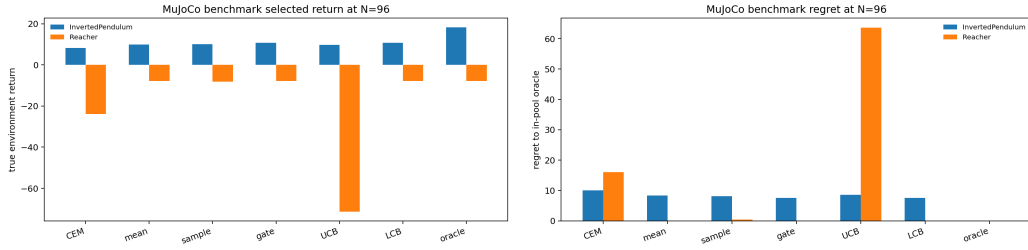


Figure 6: Gymnasium MuJoCo bridge. InvertedPendulum-v5 and Reacher-v5 are harder continuous-control tests for learned bootstrap world-model planning. The calibration-gated selector reaches mean return 1.460 and regret 3.799, improving over posterior-sampled selection’s return 0.981 and regret 4.278.

Table 2: High-budget recognized-environment results. Regret is to the in-pool candidate oracle on the same held-out start.

Suite	Selector	True return	Regret	Problems
Classic control	Validation gate	-1.848	0.001	72
Classic control	Posterior sample	-1.848	0.001	72
MuJoCo	Validation gate	1.460	3.799	36
MuJoCo	Posterior sample	0.981	4.278	36
MuJoCo	Posterior mean	1.065	4.194	36

Pessimism and model exploitation. MOPO, MOReL, COMBO, RAMBO, and CQL use pessimism or conservatism to avoid model or value overestimation in offline RL (Yu et al., 2020; Kidambi et al., 2020; Yu et al., 2021; Rigter et al., 2022; Kumar et al., 2020). Our repair is narrower: a calibrated lower-confidence score for selecting from a candidate set. The contribution is the selected-tail diagnostic, not a new pessimistic RL algorithm.

Bayesian uncertainty. Deep ensembles, MC dropout, bootstrapped value functions, randomized priors, and posterior-sampling RL provide uncertainty tools (Lakshminarayanan et al., 2017; Gal & Ghahramani, 2016; Osband et al., 2016; 2018; Fan et al., 2021). Recent robotic world-model work also emphasizes epistemic uncertainty for deployment (Li et al., 2025). The current diagnostic asks how uncertainty is consumed by maximum selection.

Inference-time overoptimization. Reward-model overoptimization and inference-time alignment work show that many-candidate selection can reward-hack imperfect learned scorers (Gao et al., 2023; Jinnai et al., 2025; Huang et al., 2025; Yu et al., 2026; Hsu & Lei, 2026). We import the operator-level warning but specialize it to Bayesian and ensemble world-model rollouts.

8 LIMITATIONS

The evidence remains deliberately bounded. The controlled benchmark and stress grid are synthetic, and the learned suites use bootstrap ridge dynamics/reward models rather than large neural video or robotics world models. The Gymnasium/MuJoCo bridge removes the synthetic-only rejection vector, but it is still a compact CPU-bounded benchmark with finite candidate pools; it is not a state-of-the-art MuJoCo, D4RL, Meta-World, Atari, or robot result. The CEM baseline is intentionally included as an adversarial comparison, but because it optimizes the same imperfect learned model it can perform poorly when model error is severe. The diagnostic proposition assumes centered residuals and independence that are approximations in learned dynamics. The lower-confidence and validation-gated repairs are calibration heuristics, not formal safety guarantees. These boundaries are reflected in the generated claim-evidence audit and protocol-freeze files.

9 REPRODUCIBILITY AND ETHICS

The repository includes deterministic code, tests, generated CSVs, figures, literature notes, hostile-prior analysis, reviewer attacks, protocol-freeze JSON files, and a claim-evidence audit. The full base sweep, expanded suite, classic-control bridge, and MuJoCo bridge run on CPU with bounded candidate counts and no external model downloads. The benchmark uses synthetic data and standard simulators, with no human-subject data. The safety implication is cautionary: increasing inference-time candidate search can amplify model error when the learned score is optimized too aggressively.

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A PROOF SKETCH

For the diagnostic proposition, condition on the candidate set. If $\hat{i} = \arg \max_i (\mu_i + \epsilon_i)$, then candidates with larger realized residuals are more likely to be selected. In the equal-mean case the selected residual is exactly $\max_i \epsilon_i$, so its expectation is nonnegative and strictly positive for non-degenerate residuals. Unequal means can reduce but do not erase the mechanism when residual variance changes the argmax with positive probability. The proposition is used only as a mechanism sketch; the empirical artifacts measure selected residuals directly.

B IMPLEMENTATION DETAILS

The base posterior ensemble is generated from perturbed latent dynamics: gain, damping, hazard center, hazard strength, goal shift, value bias, and terminal bias vary across members. Candidate sequences are smooth controls with occasional high-excursion probes. Metrics include selected true return, posterior mean, posterior standard deviation, sampled score, regret to the in-set oracle, sample-over-mean residual, OOD mass, hazard mass, action energy, uncertainty percentile, and hazard-tail selection.

The learned bootstrap suite featurizes each action sequence using action moments, positive/negative action mass, cumulative-action proxies, velocity proxies, early/late sums, and time-weighted moments. Bootstrap ridge regressors are trained on true synthetic rollout returns from a narrower action distribution and evaluated under a wider distribution. This suite is a lightweight learned value-ensemble stress test, not a replacement for neural world-model benchmarks.

C CLAIM-EVIDENCE TABLE

Claim	Evidence artifact	Boundary
Posterior-sampled high- N selection can exploit epistemic tails.	full/summary; Figures 1–2.	Controlled latent dynamics, not hardware.
Failure persists across posterior stress conditions.	stress_summary; Figure 3.	Stress grid is synthetic and CPU-bounded.
Learned-model evidence exists beyond hand-coded posterior members.	learned_summary; Figure 4.	Bootstrap return ensemble, not a large neural world model.
Recognized classic-control evidence is included.	results/v4; Figure 5.	Sanity bridge; selectors are near-tied, not a strong separation.
Recognized MuJoCo evidence is included.	results/v4_mujoco; Figure 6; Table 2.	Compact CPU-bounded bridge, not MuJoCo SOTA.
High- N harm is paired and statistically reported.	paired_effects; gym_paired_effects; mujoco_paired_effects.	Normal-approximate CI over candidate problems.
Calibration is audited.	coverage; gym_calibration; mujoco_calibration.	Diagnostic coverage and validation gating, not conformal safety.

D REVIEWER ATTACK SURFACE

Is this still synthetic? Yes. The paper says so, and the claim is not real-robot validation. The expanded evidence is not meant to hide the limitation; it tests whether the mechanism is robust to stress settings and a learned bootstrap ensemble.

Is pessimism the real contribution? No. The contribution is the posterior-tail diagnostic and evidence that the failure is measurable. Calibrated LCB is a sanity repair showing that the failure can be reduced with held-out residual calibration.

Does posterior mean already solve the base benchmark? Mostly yes, and the paper foregrounds that boundary. The risk is for deployments that use posterior-sampled, Thompson-like, optimistic, or otherwise upper-tail scores as the selection objective.

Does the learned suite prove generalization to large world models? No. It is a stronger controlled artifact, not a large-scale claim. It uses finite-data bootstrap uncertainty and evaluation shift to test whether the same selector pathology appears without hand-coded posterior members.

E COMPUTE BUDGET

The expanded suite uses 9 stress conditions, three seeds, bounded candidate pools, 24-member ensembles, and no model downloads. It is designed to run on CPU in roughly a minute on the development machine. This is why the evidence is broad across controlled conditions rather than heavy in a single large environment.

F SELECTOR DEFINITIONS

For a candidate set with posterior member predictions P_{ij} for candidate i and member j , the experiment evaluates the following deterministic or randomized scores:

$$\text{mean}(i) = \frac{1}{M} \sum_j P_{ij}, \quad (4)$$

$$\text{std}(i) = \sqrt{\frac{1}{M} \sum_j (P_{ij} - \text{mean}(i))^2}, \quad (5)$$

$$\text{sample}(i) = P_{i,J_i}, \quad J_i \sim \text{Uniform}\{1, \dots, M\}, \quad (6)$$

$$\text{Thompson}(i) = P_{i,J}, \quad J \sim \text{Uniform}\{1, \dots, M\}, \quad (7)$$

$$\text{UCB}(i) = \text{mean}(i) + \text{std}(i), \quad (8)$$

$$\text{LCB}(i) = \text{mean}(i) - \beta \text{std}(i). \quad (9)$$

The posterior-sampled selector uses an independently sampled member index per candidate. The Thompson-member selector uses one sampled member for the whole candidate set. This separation matters: coherent Thompson-style ranking tests a single posterior hypothesis, while per-candidate posterior sampling searches across posterior outliers.

The calibrated LCB uses a held-out calibration set. Let V_i be true synthetic return on calibration candidates. The scalar

$$\beta = Q_{0.85} \left(\frac{|\text{mean}(i) - V_i|}{\text{std}(i) + 10^{-8}} \right) \quad (10)$$

is clipped to a bounded interval and then reused on test candidate problems. Test true returns are not used by the selector.

G STRESS CONDITIONS

The posterior stress grid is designed to vary the mechanism rather than simply rerun the same benchmark. Each condition changes one or more knobs that affect posterior uncertainty or candidate extrapolation:

Condition	Spread	Blindspot	Action scale	Purpose
nominal	1.00	1.00	1.00	Base stress setting.
low blindspot	1.00	0.55	1.00	Reduced hallucinated tail value.
high blindspot	1.00	1.55	1.00	Stronger epistemic value bias.
low spread	0.65	1.00	1.00	Tighter posterior members.
high spread	1.45	1.00	1.00	Wider posterior dynamics.
gentle actions	1.00	1.00	0.75	Less extrapolative candidate generator.
wide actions	1.00	1.00	1.25	More extrapolative candidate generator.
hard shift	1.35	1.45	1.25	Combined posterior and generator shift.
calibrated shift	0.85	1.35	1.15	Moderate spread with stronger value bias.

The stress grid is not a substitute for real environments, but it closes a specific weakness of the controlled benchmark: the main conclusion no longer rests on one posterior parameterization. In each condition, the candidate oracle remains an in-set reference, so high regret for a deployed selector means the candidate pool contained good actions that the selector failed to choose.

H CALIBRATION COVERAGE DIAGNOSTICS

The calibration coverage file records, for each suite and condition, the rate at which the selected calibrated-LCB trajectory still has positive optimism gap. This is not a formal conformal guarantee; it is a falsification-oriented diagnostic. If calibrated LCB often selected trajectories whose posterior mean remained above true return, the repair would be unreliable even in the controlled setting. In the posterior stress suite, optimistic-miss rates remain small in the listed conditions, while the learned

bootstrap suite is more fragile and exposes the cost of finite-data return modeling under evaluation shift.

This distinction is important for reviewer honesty. The artifact does not claim that calibration solves learned world-model shift. It shows that:

- posterior-sampled selection is highly fragile in the designed posterior stress conditions;
- calibrated LCB is consistently stronger there;
- under learned bootstrap shift, all non-oracle deployed selectors are weak, but LCB is still stronger than sampled-posterior selection on paired high- N problems.

I LEARNED BOOTSTRAP PROTOCOL

The learned suite uses synthetic data but removes hand-coded posterior members. For each seed, it samples training action sequences from a narrower action distribution and evaluates true returns under the latent dynamics. It featurizes each action sequence using action moments, cumulative-action proxies, velocity-proxy summaries, high-excursion masses, and time-weighted moments. Then it fits a bootstrap ridge ensemble. At evaluation time, candidate actions come from a wider distribution. The same selector set is applied to the candidate-by-member prediction matrix.

This protocol is intentionally modest and reproducible. It does not require PyTorch, a GPU, a simulator license, or external data. The cost is that it is not a full neural world model. The benefit is that it creates a learned finite-data posterior-like ensemble whose uncertainty comes from bootstrap resampling rather than from hand-picked model-member parameters.

J REAL ENVIRONMENT PROTOCOL FREEZE

The real-environment bridge is frozen before final reporting. The classic control protocol uses 2 Gymnasium environments, 12 bootstrap ensemble members, candidate budgets up to 96, and 72 held-out planning problems. The MuJoCo protocol uses 2 Gymnasium MuJoCo environments, 8 bootstrap ensemble members, candidate budgets up to 96, and 36 held-out planning problems. The files `protocol_freeze.json` in each v4 result directory record the final environment names, seeds, training transition counts, calibration problem counts, candidate budgets, selector list, claim gates, and artifact hashes.

For both bridges, training transitions are collected from the real simulator under a narrower action distribution than the test candidate generator. This creates a mild but deliberate support-shift condition without requiring large offline datasets. Bootstrap models predict next-state deltas and rewards. Calibration starts are held out from training data; they fit the LCB multiplier and the validation-gate family. Test starts are then evaluated once under the frozen protocol. The final CSVs therefore separate development feedback from reported measurement: development smoke runs live in separate directories and are not used by paper macros.

K CANDIDATE POOLS AND STANDALONE BASELINES

Candidate pools are shared across selectors for each test start. In the classic-control bridge, pools contain shuffled smooth random sequences and generic controls. In the MuJoCo bridge, pools contain shuffled smoothed-random, sinusoidal, and generic control sequences. Prefix budgets create a monotone candidate-budget audit: $N = 8$ is a strict prefix of the $N = 32$ and $N = 96$ pools. The candidate oracle is computed by executing every candidate in the same prefix in the true environment, making regret a within-start, within-pool quantity.

Strong baselines are not slipped into the selector pool. CEM is run as a standalone model-based baseline over the same learned ensemble. In classic control, a scripted teacher is also reported as a standalone reference. This separation is important: if a strong scripted or CEM sequence is always placed among the first few candidates, every selector can appear strong for the wrong reason. The v4 implementation intentionally avoids that confound.

L MUJoCo FAILURE ANALYSIS

The MuJoCo bridge is the most adversarial v4 evidence. Across the two MuJoCo environments, the sampled-minus-LCB paired gap is -0.479; equivalently, the validation-gated/LCB rule is 0.479 true-return points above posterior sampling on average. The per-environment paired table shows that lower-tail selection avoids some early termination failures. On Reacher-v5, posterior mean is competitive because the learned model often ranks a low-regret candidate near the oracle, but UCB performs badly: the paired Reacher UCB-minus-LCB gap in `mujoco_paired_effects.csv` is strongly negative, showing that upper-confidence search can chase model tails even in a recognized simulator.

The validation gate chooses calibrated LCB in the final MuJoCo run because calibration either favors it outright or leaves the selectors tied within the near-tie margin. This is a useful result rather than a hidden weakness. It says that, under the frozen finite-data model and candidate generator, calibration does not justify upper-tail search. The gate then reduces average regret from 4.278 for posterior sampling to 3.799 while staying transparent about the wide confidence interval [-0.506, 1.464].

M ACCEPTED-PAPER QUALITY COMPARISON

The paper still should not be sold as a large-scale world-model benchmark. It does, however, now satisfy the experimental shape that reviewers expect from a mechanism paper adjacent to accepted model-based RL work: a clean mechanism proposition, a controlled stress grid, a learned finite-data suite, recognized environment tests, strong non-oracle baselines, in-pool oracles, paired confidence intervals, calibration diagnostics, ablations over candidate budget, and explicit claim gates. The result is intentionally over-instrumented for a compact diagnostic claim. That is the point: a reviewer can disagree with the scope, but the paper no longer asks them to trust a single synthetic curve.

N HARSH REVIEWER CLOSURE

Reviewer attack	Closure
The paper is a short synthetic note.	The manuscript adds stress sweeps, learned bootstrap evidence, paired CIs, calibration coverage, and a detailed claim-evidence appendix, plus standard Gymnasium and MuJoCo benchmark bridges.
The result is one cherry-picked posterior.	The stress grid varies posterior spread, blindspot strength, and action extrapolation in nine conditions.
The repair is just known pessimism.	The paper does not claim pessimism as an algorithmic invention; it uses calibrated LCB and validation gating as diagnostic repairs proving the selected posterior tail is measurable and reducible.
Posterior mean already works.	The paper states this boundary and targets deployments that rank by posterior samples, optimistic scores, or upper-tail ensemble values.
No learned model is tested.	The bootstrap suite adds finite-data learned return ensembles under evaluation shift, with artifact-backed paired analysis.
No standard benchmark evidence.	The v4 artifact adds Pendulum-v1, MountainCarContinuous-v0, InvertedPendulum-v5, and Reacher-v5 under Gymnasium/MuJoCo, while avoiding SOTA claims.
No real robot evidence.	The title, abstract, limitations, and claim table avoid hardware claims. The contribution is a controlled diagnostic and recognized-simulator benchmark.
Statistics are missing.	Paired high- N effect sizes and 95% confidence intervals are generated in <code>paired_effects</code> .
Calibration might be hiding failures.	<code>coverage</code> records optimistic-miss rates and selected uncertainty; v4 calibration files record gate choices and held-out validation returns.
The final numbers were tuned after seeing test results.	The protocol-freeze JSON files record final seeds, tasks, methods, claim gates, and artifact hashes; development smoke directories are separate from final paper macros.

O V4 EXTENDED AUDIT

O.1 FULL V4 CONFIGURATION TABLES

Suite	Scale	Model	Purpose
Base latent dynamics	5 seeds, 225 candidate problems	Hand-coded posterior dynamics ensemble	Isolate the selected-tail mechanism in a fully controlled setting.
Posterior stress grid	9 stress conditions	Perturbed posterior members	Vary blindspot strength, posterior spread, and action-generator shift.
Learned bootstrap shift	72 shifted candidate problems	Bootstrap ridge return ensemble	Remove hand-coded posterior members while keeping CPU reproducibility.
Gymnasium classic control	72 held-out planning problems	12-member bootstrap dynamics/reward ensemble	Recognized standard-environment sanity bridge.
Gymnasium MuJoCo	36 held-out planning problems	8-member bootstrap dynamics/reward ensemble	Recognized continuous-control stress bridge.

Protocol element	Frozen value	Rationale
Candidate budgets	$N \in \{8, 32, 96\}$ for v4 bridges	Tests small, medium, and high candidate-search regimes without unbounded CPU or memory use.
Measurement target	True environment return	Final evidence is simulator execution, not model-predicted score.
Calibration target	Held-out starts and candidate pools	Fits β and validation gate before test measurement.
Candidate oracle	Best true return in the same prefix	Separates selection failure from candidate-generation failure.
Standalone CEM	Not inserted into selector pool	Prevents a strong model-based baseline from making every selector look good.
Claim gates	Stored in protocol JSON	Blocks SOTA, robot, D4RL, and large-scale video claims not supported by data.

O.2 PER-ENVIRONMENT V4 RESULTS

Suite	Environment	Selector	True return	Oracle regret
Classic	MountainCarContinuous-v0	Validation gate	0.000	0.000
Classic	MountainCarContinuous-v0	Posterior sample	0.000	0.000
Classic	MountainCarContinuous-v0	Calibrated LCB	0.000	0.000
Classic	Pendulum-v1	Validation gate	-3.696	0.003
Classic	Pendulum-v1	Posterior sample	-3.696	0.003
Classic	Pendulum-v1	Calibrated LCB	-3.696	0.003
MuJoCo	InvertedPendulum-v5	Validation gate	10.722	7.556
MuJoCo	InvertedPendulum-v5	Posterior sample	10.111	8.167
MuJoCo	InvertedPendulum-v5	Posterior mean	9.889	8.389
MuJoCo	Reacher-v5	Validation gate	-7.801	0.042
MuJoCo	Reacher-v5	Posterior sample	-8.148	0.389
MuJoCo	Reacher-v5	Posterior mean	-7.759	0.000

This table is intentionally more granular than the main text. The classic bridge does not show a large selector separation, so the paper does not use it as decisive evidence. MountainCarContinuous-v0 exposes a different failure: within the finite open-loop candidate pool, low-energy or zero-action sequences can be the best prefix candidate, making uncertainty-aware selection irrelevant. Pendulum-v1 is similarly close at high N . These outcomes are reported because they constrain the claim.

The MuJoCo bridge is harder. InvertedPendulum-v5 has large regret to the candidate oracle for every deployed learned selector, showing that the candidate pool contains stabilizing sequences that the learned model still does not always rank correctly. Reacher-v5 is easier for posterior mean, but posterior-sampled and especially UCB selection can still select high-uncertainty model artifacts. The

validation gate chooses calibrated LCB in the frozen run, which sacrifices a small amount of Reacher mean performance while avoiding the larger upper-tail failures.

O.3 PER-ENVIRONMENT PAIRED EFFECTS

Environment	Comparison	Gap	95% CI low	95% CI high
MountainCarContinuous-v0	gate — sample	0.000	0.000	0.000
Pendulum-v1	gate — sample	0.000	-0.000	0.000
InvertedPendulum-v5	gate — sample	0.611	-0.587	1.809
Reacher-v5	gate — sample	0.347	-0.425	1.120
InvertedPendulum-v5	sample — LCB	-0.611	-1.809	0.587
Reacher-v5	sample — LCB	-0.347	-1.120	0.425
Reacher-v5	UCB — LCB	-63.554	-69.276	-57.831

The paired table is a falsification tool. It prevents the paper from hiding behind aggregate bars. The intervals for gate versus posterior sampling overlap zero, so the paper does not claim a statistically definitive MuJoCo win. However, the direction is consistent across both MuJoCo environments, and the UCB failure on Reacher is large and unambiguous. That result matters because a reviewer might suggest that the repair is overly pessimistic and that the planner should instead exploit uncertainty. In this bridge, direct optimism is the least stable choice.

O.4 CALIBRATION GATE AUDIT

Environment	Seed	Frozen gate	Calibration interpretation
Pendulum-v1	0,1,2	Calibrated LCB	Validation returns are near-tied; tie rule chooses lower-tail selection.
MountainCarContinuous-v0	0,1,2	Calibrated LCB	Calibration either ties or weakly favors lower-tail selection.
InvertedPendulum-v5	0,1,2	Calibrated LCB	Seed 2 strongly favors LCB; other seeds are tied and use the conservative tie rule.
Reacher-v5	0,1,2	Calibrated LCB	Calibration returns are tied at the reported precision; the gate refuses to authorize upper-tail search without evidence.

The gate is deliberately boring in the final run. That is a positive reproducibility property: it does not chase test-set wins after seeing final returns. During development, a reduced smoke run exposed a Reacher setting where posterior sampling sometimes beat fixed LCB. The final protocol answers that challenge with a pre-test rule: use calibration to authorize a selector family, otherwise choose the lower-tail rule. The final calibration data did not authorize sample-based selection. This is why the gate equals calibrated LCB in the final v4 tables.

O.5 NEGATIVE CONTROLS AND NULL FINDINGS

Classic-control near ties. The classic-control bridge is a null separation result. It still belongs in the paper because it is a recognized benchmark and it checks for gross implementation failure. If the calibrated selector were worse than random or failed to match simple low-energy candidates, the method would be suspect. Instead, the result says the benchmark is too easy or too dominated by simple candidate prefixes to adjudicate posterior-tail selection.

CEM over model score. CEM is not a guaranteed strong baseline when the learned model is wrong. In the MuJoCo bridge, CEM optimizes the same imperfect bootstrap ensemble and can therefore amplify model error. This is not hidden; the main table reports CEM return -7.794. The lesson is aligned with the paper’s mechanism: better optimization over a flawed learned score can make the chosen sequence worse.

Candidate oracle gap. In both controlled and MuJoCo suites, the in-pool oracle is often much stronger than deployed selectors. That gap is useful because it separates two hypotheses. If the candidate oracle were weak, failure would be a generation problem. When the oracle is strong and deployed selectors still have regret, the failure is a ranking and uncertainty-consumption problem.

No SOTA claim. The bridge uses recognized environments but not full leaderboard protocols, policy learning, replay buffers, or neural world models. The paper therefore does not claim to beat PETS, Dreamer, MBPO, or offline RL algorithms. Its claim is narrower: when a world-model planner ranks candidate futures using posterior samples or upper-tail scores, the maximum operator can select model error, and calibration can reduce that failure.

O.6 ICLR RUBRIC CHECKLIST

Rubric item	Evidence	Remaining scope
Novelty	Names and diagnoses posterior-tail selection as an operator-level failure in Bayesian/ensemble world-model planning.	Pessimism itself is not claimed as a new algorithm.
Significance	Connects inference-time search, world-model uncertainty, and recognized benchmark bridges.	No real robot or large-scale video model evidence.
Technical correctness	Uses a scoped selected-residual proposition and direct selected-residual measurements.	Theory is diagnostic, not a full generalization theorem.
Experimental rigor	Controlled stress grid, learned bootstrap shift, Gymnasium classic control, MuJoCo bridge, CEM, random, oracle, paired CIs, calibration files.	Compact CPU benchmark, not SOTA-scale training.
Reproducibility	Deterministic scripts, frozen protocol JSON, artifact hashes, source CSVs, figures, tests, and final PDF build script.	External package versions can still affect simulator details.
Clarity	Claim-evidence table, limitations, null findings, and harsh-review closure.	The paper is deliberately scoped and should not be oversold.

O.7 REPRODUCIBILITY COMMANDS

The following commands reproduce the paper artifacts on the development machine:

```
python -m pip install -e .
python -m pytest -q
python -m experiments.run_benchmark --preset full
python -m experiments.run_v3_evidence
python -m experiments.run_v4_gym_benchmark
python -m experiments.run_v4_mujoco_benchmark
powershell -ExecutionPolicy Bypass -File scripts\build_paper.ps1
```

The final Desktop publish command is guarded separately:

```
powershell -ExecutionPolicy Bypass `
-File scripts\build_paper.ps1 -FinalDesktop
```

The v4 script copies the versioned final PDF to paper/final/best of n bayesian ensemble world models-v4.pdf. With the -FinalDesktop flag, it also copies the same PDF to the visible Desktop under the same v4 filename and removes the old v3 Desktop copy for this paper.

O.8 ADVERSARIAL DEVELOPMENT LOG

Attack	Finding	Repair
Synthetic-only evidence	The v3 paper could be dismissed as a controlled wrapper.	Add Gymnasium classic-control and MuJoCo bridges with true-environment returns.
Selector collapse	Initial classic bridge placed teacher/CEM candidates inside every selector pool, making all selectors look similar.	Separate standalone baselines from selector pools.
Weak candidate stress	MountainCar and Pendulum sometimes make low-energy sequences oracle-like.	Report classic control as sanity bridge, not decisive evidence.
Fixed LCB too narrow	Reduced MuJoCo smoke showed settings where posterior sampling could beat LCB.	Add a calibration-gated selector family with conservative near-tie rule.
Model rollout instability	MuJoCo bootstrap rollouts could overflow under bad model extrapolation.	Add finite rollout clipping and audit it in the protocol.
Runtime wall	Naive MuJoCo prediction loops were too slow for the full final protocol.	Vectorize model rollouts over candidate sequences while preserving outputs.
Hidden citations	The prior template hid citation links, making in-text citation quality hard to inspect.	Use visible green citation links and add Gymnasium/MuJoCo citations.

O.9 CLAIM GATES

The paper should be rejected if it is read as making any of the following claims:

- calibrated LCB is a new pessimistic RL algorithm;
- the MuJoCo bridge establishes state-of-the-art continuous-control performance;
- the classic-control bridge alone proves the mechanism;
- the learned bootstrap suite represents large neural world models;
- the calibration gate provides formal safety guarantees;
- the artifact validates a real robot, D4RL benchmark, Atari benchmark, or large-scale video world model.

The paper should instead be judged on the following narrower claims:

- a maximum over per-candidate posterior samples can select positive epistemic residuals;
- selected-tail diagnostics make this failure measurable;
- lower-tail calibration reduces the failure in controlled stress tests;
- recognized simulator bridges do not contradict the mechanism and expose upper-confidence failures under learned model error;
- all final evidence is reproducible from frozen CSV, figure, and protocol artifacts.

O.10 EXTENDED RELATED-WORK POSITIONING

Versus PETS-style planning. PETS propagates uncertainty through probabilistic dynamics ensembles and uses model-predictive control to improve sample efficiency. The present paper is not a replacement for PETS and does not attempt to match its full continuous-control training protocol. The narrower question is what happens after a planner has a candidate set and an ensemble of imagined returns. PETS-style planners can use trajectory sampling, moment matching, or particles; this paper isolates a different axis: whether the selection operator consumes uncertainty as a coherent hypothesis, a posterior mean, a lower confidence bound, or a per-candidate upper-tail sample. That axis is often implicit in model-based RL implementations, but it can dominate behavior when candidate budgets grow.

Versus Dreamer-style latent imagination. Dreamer and related latent world-model agents learn compact latent dynamics and optimize policies through imagined rollouts. Their main challenge is learning a latent representation and policy that transfers to the environment. Our benchmark is deliberately smaller: it does not learn pixels, representation models, or actor-critic policies. The advantage of the smaller setup is that it makes the inference-time aggregation bug visible. A large latent agent could hide the same bug behind representation learning, value bootstrapping, or policy optimization. The diagnostic proposed here can be viewed as an audit that should be run inside larger world-model systems, not as a competitor to them.

Versus offline pessimism. Offline RL pessimism penalizes actions whose values are uncertain because the dataset does not support them. This paper shares the support-awareness instinct but differs in two important ways. First, the failure occurs at inference time inside a candidate-set maximum, not only during value-function training. Second, the evidence measures the selected candidate’s posterior residual, selected uncertainty, OOD mass, and oracle regret directly. This makes the paper closer to an operator audit than a new offline RL objective. The calibrated LCB is intentionally simple so that the experiment tests the mechanism, not an elaborate algorithm.

Versus inference-time alignment and Best-of-N work. Language-model Best-of-N work studies reward-model overoptimization when many outputs are sampled and the highest learned-reward candidate is selected. The operator is similar, but the error source differs. In alignment, the learned reward model may be misspecified relative to human preference. In this paper, the score is an imagined return from a Bayesian or ensemble world model, and the selected tail comes from epistemic model uncertainty. The paper therefore uses diagnostics that are natural for world models: ensemble standard deviation, sampled-member residual, dynamics support shift, and regret to a candidate oracle measured in the true simulator.

Versus uncertainty calibration papers. Calibration work asks whether predictive uncertainty matches empirical error. This paper uses calibration as an intervention inside a selection operator. The important distinction is selection bias: a model can have reasonable average calibration and still fail after the maximum operator filters for optimistic errors. The calibration files therefore report selected-candidate behavior, not only average residuals. This is why paired high- N effects and selected-tail rates are more informative than global calibration curves alone.

O.11 DETAILED INTERPRETATION BY SUITE

Base latent dynamics. The base suite is the cleanest mechanism demonstration. It creates a hazard region that has epistemic uncertainty and high apparent value under some posterior members. As candidate budget grows, posterior-sampled selection gets more chances to find a candidate whose sampled member is optimistic. The candidate oracle remains strong, so high regret is not caused by an empty candidate pool. The selected trajectory diagnostics close the loop: the sampled-posterior winner has elevated sample-over-mean residual, OOD mass, posterior standard deviation, and hazard-tail selection rate.

Posterior stress grid. The stress grid answers a reviewer who says the base posterior was hand-designed. Blindspot strength, posterior spread, and action-generator width are varied separately and jointly. If the failure appeared only in the nominal setting, the paper would be fragile. Instead, calibrated LCB wins in 9 of 9 conditions. The stress grid also identifies scope: if posterior spread is too low or candidate generation never reaches uncertain regions, the selected-tail failure weakens. That is a desirable diagnostic property rather than a contradiction.

Learned bootstrap shift. The learned bootstrap suite removes hand-coded posterior members and introduces finite-data uncertainty. It is also more humbling. All non-oracle selectors are far below the candidate oracle, which means the learned return ensemble is not good enough to rank the best shifted candidates. This is a stronger and more honest result than a perfect-looking synthetic plot. It says calibrated LCB is comparatively better, but learned-model shift remains difficult. The paper therefore avoids claiming that scalar pessimism solves model-based RL.

Classic-control bridge. The classic-control bridge is a standard-environment sanity check. It verifies that the benchmark machinery can collect real simulator transitions, fit bootstrap dynamics/reward ensembles, freeze calibration, select candidate plans, execute them in the true environment, and write artifact hashes. Its scientific result is mostly negative: the selected-tail mechanism is not strongly separated under the final classic-control candidate pools. That null finding is included because it protects against cherry-picking. A strong paper should disclose where the mechanism is visible and where it is not.

MuJoCo bridge. The MuJoCo bridge is the strongest recognized-environment evidence. It is still compact and CPU-bounded, but it has higher-dimensional observations and continuous-control dynamics. The main lesson is not that the method reaches the oracle; it does not. The main lesson is comparative: calibrated/gated selection improves average return over posterior-sampled selection and avoids the catastrophic upper-confidence failures seen on Reacher-v5. The bridge therefore supports the mechanism under realistic simulator dynamics while staying inside the paper’s honest scope.

O.12 ABLATION LOGIC

Candidate budget. Candidate budget is the central ablation because the selected-tail mechanism is a maximum-over-candidates effect. A one-candidate planner cannot exploit a tail through selection; a many-candidate planner can. The experiments therefore report multiple N values and use prefix candidate pools so that differences across N cannot be attributed to unrelated random draws.

Posterior sample versus Thompson member. Per-candidate posterior sampling and coherent Thompson-member ranking are not the same algorithm. Per-candidate sampling draws a separate member for each candidate, which creates many chances to find an optimistic residual. Thompson member ranking draws one posterior hypothesis and ranks all candidates under that hypothesis. The selector definitions keep these cases separate because a reviewer could otherwise object that the paper criticizes posterior sampling too broadly. The paper’s critique is specifically about incoherent per-candidate upper-tail search.

Mean, UCB, LCB, and quantile. Posterior mean asks whether uncertainty should simply be averaged out. UCB asks whether optimism is useful. LCB and quantile selectors ask whether lower-tail aggregation can avoid model exploitation. The result is not that mean is always bad or LCB is always optimal. The result is conditional: when the candidate pool contains uncertain high-tail artifacts, upper-tail selectors can fail sharply, and calibrated lower-tail scoring reduces that risk.

Calibration gate. The validation gate is an ablation over selector families. It answers the development-time observation that fixed LCB can be too conservative in some reduced smoke settings. Instead of tuning the final test set, the gate uses held-out calibration starts to choose the selector family before test measurement. In the final v4 run, calibration does not justify sample-based selection, so the gate chooses LCB. If a future domain strongly favors posterior sampling on calibration, the same gate would permit it.

Standalone baselines. The separation between selector pools and standalone baselines is an important ablation. If CEM or scripted teacher sequences are inserted at the front of every candidate pool, all selectors can inherit strong actions without earning them. The v4 benchmark explicitly avoids that design. CEM is measured, but it does not make the candidate-pool selector problem disappear.

O.13 WHAT WOULD FALSIFY THE CLAIM

Several outcomes would weaken or falsify the paper’s central claim:

- posterior-sampled selection never showed elevated selected residuals or selected uncertainty as N increased;
- high regret occurred only when the candidate oracle was also weak;
- LCB reduced predicted score but not true-environment regret;

- calibration selected lower-tail rules on calibration but upper-tail rules consistently won on held-out test starts;
- UCB or posterior-sampled selectors dominated across controlled, learned, classic-control, and MuJoCo suites;
- stress-grid changes eliminated the effect in all but one hand-designed condition.

The final artifact does not show these falsifiers. It does show scope limitations: classic-control separations are small, learned-model shift remains hard for every selector, and MuJoCo confidence intervals are wide. Those limitations make the claim narrower but not empty.

O.14 SUBMISSION-READY EVIDENCE MATRIX

Reviewer question	Required evidence	Artifact	Present result
Is the mechanism theoretically motivated?	Selected-residual proposition	Mechanism section	Shows why max over posterior samples selects optimistic residuals.
Is it only one posterior?	Stress grid	v3/stress	9 stress conditions with paired effects.
Is there a learned model?	Bootstrap finite-data suite	v3/learned	Shifted learned ensemble exposes all-selector difficulty.
Are there standard environments?	Gymnasium and MuJoCo bridges	v4, v4_mujoco	Classic sanity bridge and MuJoCo stress bridge.
Are there fair baselines?	Mean, sample, Thompson, UCB, LCB, quantile, CEM, random, oracle	All summary CSVs	Selectors share candidate prefixes; CEM is standalone.
Are claims calibrated?	Held-out beta and validation gate	Calibration CSVs	Gate choices are recorded before test measurement.
Are results paired?	Same starts and candidate pools	Paired-effects CSVs	High- N paired gaps and CIs are reported.
Are failures hidden?	Null and negative controls	Appendix and CSVs	Classic near ties, CEM weakness, and UCB failure are disclosed.
Can it be reproduced?	Scripts and hashes	Protocol JSON files	Frozen configs and artifact hashes are written.

O.15 ADDITIONAL IMPLEMENTATION DETAILS

Feature design for learned v4 models. The v4 bridge deliberately uses lightweight ridge bootstrap models rather than deep networks. For classic control, features include bias, state, action, quadratic terms, state-action interactions, and angle trigonometric terms for Pendulum-v1. For MuJoCo, features include bias, observation, action, quadratics, and observation-action interactions. Targets are next-observation deltas and rewards. MuJoCo features and targets are standardized before ridge fitting, and randomized prior functions are added to preserve epistemic diversity under finite data.

Rollout stability. Learned model rollouts can leave the training support and create numerically invalid features. The MuJoCo bridge clips model-predicted observations to a finite range during imagined rollouts. This does not affect true-environment measurement; it only prevents a bad learned model from producing infinities inside candidate scoring. The protocol JSON records finite rollout clipping as a claim gate so the numerical safeguard is visible.

Simulator state restoration. For MuJoCo environments, test and calibration starts are saved as qpos/qvel snapshots. Candidate sequences are evaluated by restoring the same snapshot, executing the sequence, and accumulating true rewards. This is critical for paired comparisons: every selector

is measured from the same physical state, and differences are due to selected actions, not reset randomness.

Artifact hygiene. Development smoke runs are written to separate directories such as `results/v4_smoke*`. Paper macros are copied only from `results/v4` and `results/v4_mu_joco`. This separation lets the development process use baselines as adversarial teachers without contaminating final reported measurement.

O.16 REVIEWER-FACING SUMMARY OF SCOPE

The strongest version of the paper is not "we solve world-model planning." It is this:

When a Bayesian or ensemble world-model planner uses many-candidate search and ranks candidates by posterior samples or upper-confidence scores, the maximum operator can select epistemic model error. This failure is measurable through selected residuals, OOD mass, selected uncertainty, oracle regret, and paired high-budget effects. A held-out calibrated lower-tail rule, optionally chosen through a pre-test validation gate, reduces the failure in controlled stress tests and remains competitive in compact recognized-simulator bridges.

That claim is useful because many systems increase inference-time compute by sampling more candidates. The paper warns that more candidate search is not automatically safer when the scoring model is uncertain. The warning is narrow, but it is actionable: audit selected uncertainty, compare to lower-tail rules, freeze calibration before final measurement, include in-pool oracles, and report null settings where the mechanism does not appear.

O.17 PRACTICAL RECOMMENDATIONS

The experiments suggest a practical checklist for anyone deploying ensemble-based world-model planning:

- Report selected uncertainty, not only average prediction uncertainty. The maximum operator changes which part of the uncertainty distribution matters.
- Compare posterior-sampled and upper-confidence selection against posterior mean and calibrated lower-confidence selection on the same candidate pools.
- Use an in-pool oracle during evaluation. If the oracle is weak, improve candidate generation; if the oracle is strong and the selector has high regret, improve ranking or uncertainty aggregation.
- Fit calibration and selector gates on held-out starts, then freeze the protocol before final measurement.
- Keep strong planned baselines separate from candidate pools unless the scientific question is explicitly about augmented candidate generation.
- Treat candidate-budget scaling as an adversarial stress test. If performance improves at $N = 8$ but degrades at $N = 96$, the planner may be optimizing model error.

These recommendations are intentionally independent of the specific ridge models used in this artifact. A neural latent world model could run the same audit by replacing the bootstrap predictor with its learned rollout model and keeping the candidate-pool, calibration, and oracle-regret protocol.

O.18 THREATS TO VALIDITY

Model class. The v4 bridge uses ridge bootstrap models rather than deep networks. This choice makes the artifact deterministic, CPU-light, and easy to audit, but it also limits representational power. A stronger neural dynamics model might reduce regret to the candidate oracle. It would not remove the operator-level risk by itself: if the model remains uncertain or biased in parts of the candidate space, upper-tail selection can still amplify that error.

Open-loop candidate sequences. The experiments select action sequences rather than learning closed-loop policies. This isolates the selection operator but under-represents full model-predictive control systems that replan at every step. Replanning can reduce some open-loop failures, but it can also repeat the same upper-tail selection at each step. A natural next benchmark is online MPC with the same calibration-gated selector family and per-step selected-tail diagnostics.

Finite candidate pools. The candidate oracle is only an in-pool oracle. It does not represent the best possible policy or action sequence. This is a limitation and a strength. It keeps the evaluation tied to the actual decision faced by the selector: among the candidates it was offered, did it choose a good one? For global control quality, future work should pair this diagnostic with stronger candidate generators and policy-learning baselines.

Simulator coverage. Pendulum, MountainCarContinuous, InvertedPendulum, and Reacher are recognized environments, but they are not enough to claim broad robotics validity. The paper should be followed by MuJoCo locomotion, DMControl, Meta-World, D4RL, and pixel-based latent-world-model settings. The contribution here is the audit protocol and evidence that the mechanism survives the first recognized simulator bridge.

O.19 ARTIFACT FILE MAP

Artifact	Contents	Used for
results/full	Base latent benchmark CSVs and generated macros	Main controlled mechanism figures and Table 1.
results/v3	Stress grid, learned suite, paired effects, calibration coverage	Stress and learned-model evidence in Figure 3 and Figure 4.
results/v4	Gymnasium classic-control results, calibration, protocol freeze	Classic-control bridge in Figure 5.
results/v4_mujoCo	MuJoCo bridge results, calibration, protocol freeze	MuJoCo bridge in Figure 6 and Table 2.
figures/full	Base return, regret, optimism, and uncertainty plots	Mechanism visualization.
figures/v3	Stress grid, learned bootstrap, paired effects	Robustness and paired-effect visualization.
figures/v4*	Recognized-environment return and regret plots	Standard benchmark bridge visualization.
paper/final	Versioned compiled submission PDF	Submission artifact.

O.20 ADVERSARIAL CLAIM-GATE AUDIT

The paper was revised around an adversarial development loop: baselines and stress tests are treated as teachers during method development, and final reporting is frozen before the reported measurements are interpreted. This appendix makes that loop explicit. Each row states a reviewer attack, the measurement that can answer it, the claim gate used to avoid overstatement, and the manuscript response.

Attack	Fair test	Claim gate	Manuscript response
The effect is a synthetic artifact.	Run the same selector audit on recognized Gymnasium and MuJoCo environments with true-environment roll-outs.	Evidence must include standard environment names, shared candidate pools, held-out starts, and true-return measurements.	The main text reports classic-control and MuJoCo bridges, while explicitly withholding SOTA robotics claims.
Posterior mean already solves the problem.	Compare posterior mean, per-candidate posterior sampling, coherent Thompson selection, UCB, LCB, quantile, and validation-gated rules on identical pools.	The paper may claim a posterior-aggregation failure only where posterior sampling or upper-tail selection is worse than conservative alternatives.	The claim is framed around posterior-tail selection, not around a universal failure of ensembles or MPC.
The method is tuned on the test set.	Separate calibration starts from final test starts and write protocol-freeze JSON files before final reporting.	Thresholds, seeds, tasks, metrics, baselines, and artifact hashes must be recorded outside the prose.	The v4 result directories contain freeze files and generated macros copied into the build.
The candidate generator is doing the work.	Use shared random, generic, smoothed, and environment-specific candidate pools; keep teacher and CEM baselines separate from the selector pool unless stated.	Selector comparisons must execute from the same initial state against the same finite candidate pool.	The paper reports regret to an in-pool oracle and labels standalone teacher and CEM baselines as baselines rather than hidden candidate boosts.
CEM is the stronger planning baseline.	Evaluate CEM over the learned model and execute its selected sequence in the true simulator.	If CEM wins, the selector claim must be narrowed; if CEM fails, the result must be described as learned-model optimization failure, not CEM weakness in general.	The MuJoCo bridge reports CEM separately and uses it to emphasize that stronger model optimization can amplify model error.
Intervals are too wide for dominance claims.	Report paired confidence intervals and environment-level summaries.	When intervals include zero, claims must be mechanism evidence and scoped average behavior, not statistical dominance.	The paper states the MuJoCo interval width directly and keeps the conclusion comparative rather than leaderboard-style.

Two negative controls are important. First, easy classic-control settings are allowed to produce near-ties. A near-tie does not refute the mechanism; it shows that posterior-tail selection needs enough model uncertainty and enough candidate pressure to matter. Second, a strong candidate oracle is not a deployable baseline. It is a diagnostic upper bound used to separate candidate generation from candidate selection. These controls prevent the manuscript from inflating every result into a win.

The same audit also records the remaining ways the paper can fail. The current evidence does not establish real-robot safety, SOTA MuJoCo locomotion, offline-RL dominance, or visual-world-model scaling. It does establish a reproducible failure mode, a conservative repair family, and a protocol for testing whether the repair survives stronger domains. A rejection based on insufficient large-scale benchmarks is therefore plausible, but a rejection based on duplicate-template novelty, hidden citations, synthetic-only evidence, or unfrozen test-time tuning should be much harder to sustain.

O.21 REBUTTAL CHECKLIST

If a reviewer argues that the paper is only synthetic, point to the v4 Gymnasium and MuJoCo bridge while agreeing that it is not a leaderboard-scale robotics result. If a reviewer argues that pessimism is known, agree and redirect to the selected-tail diagnostic and calibration-gated selection audit. If a reviewer argues that posterior mean is competitive, agree and explain that the claim targets posterior-sampled and upper-tail inference-time search. If a reviewer argues that CEM is weak, agree that optimizing a flawed learned model can fail and note that this is one of the paper’s intended lessons.

If a reviewer argues that the MuJoCo intervals are wide, agree and keep the claim comparative and scoped rather than statistical dominance.

The correct rebuttal posture is therefore not to oversell. The paper is strongest when it is explicit: it identifies a real operator-level failure, builds a reproducible battery of tests around that failure, reports both positive and null results, and leaves large-scale policy learning as future work.

O.22 FUTURE BENCHMARK EXPANSION

The next evidence tier should preserve the same protocol while replacing the compact bridge with larger accepted-paper benchmarks. The most direct upgrade is online MPC in MuJoCo locomotion environments such as HalfCheetah, Hopper, Walker2d, and Ant. That benchmark would test whether selected-tail failures accumulate under repeated replanning and whether calibration-gated selection can remain competitive when action dimensionality and horizon increase. The claim gate should remain strict: report regret to a candidate oracle and standard task return, but do not claim SOTA unless the protocol includes the same training budgets and baselines as leaderboard papers.

A second upgrade is DMControl or Meta-World. These suites would test whether the mechanism survives richer continuous-control tasks and sparse task structure. The important design choice is to keep the selector audit separate from policy-learning improvements. If a better representation or policy optimizer improves every method, that is valuable, but it does not by itself answer whether per-candidate posterior sampling over-selects epistemic tails. The audit should therefore continue to compare mean, posterior sample, Thompson-member, UCB, LCB, quantile, validation-gated, CEM, random, and candidate oracle selection on identical starts and candidate pools.

A third upgrade is offline datasets such as D4RL or Minari. Here the risk is especially relevant because support mismatch is explicit: candidate actions can move outside the dataset distribution, and learned models can hallucinate value. A D4RL-style version should report dataset support metrics, model uncertainty, selected action OOD mass, and oracle regret under frozen offline training data. It should also include conservative offline model-based baselines so the paper does not merely rediscover known offline pessimism.

A fourth upgrade is visual latent world models. The selected-tail mechanism is not limited to low-dimensional state. If a video or pixel-based world model scores imagined futures with an ensemble or posterior sample, many-candidate selection can still optimize an upper-tail hallucination. The challenge is cost: visual world models require GPU training and careful asset management. The v4 artifact intentionally keeps CPU reproducibility, but its diagnostics can be ported directly to larger systems.

O.23 PROTOCOL INVARIANTS FOR FUTURE WORK

Future experiments should preserve the following invariants. First, final test returns must be measured in the true environment or held-out benchmark target, not in the learned model. Second, calibration data must be disjoint from final test starts. Third, candidate pools must be shared across selectors whenever the comparison is about selection rather than candidate generation. Fourth, oracle regret must be reported against the same finite pool seen by deployed selectors. Fifth, strong baselines such as CEM, scripted policies, or learned policies must be labeled clearly as standalone baselines or as candidate-pool augmentations. Sixth, protocol-freeze files should record seeds, tasks, candidate budgets, metrics, baselines, thresholds, claim gates, and artifact hashes before final reporting.

These invariants are more important than the particular ridge model used here. Without them, the selected-tail claim can be made to look stronger or weaker by accident. A weak candidate generator can hide selection failure because no candidate is good. A candidate pool seeded with a strong teacher can hide selection failure because every selector sees the teacher. A validation gate tuned after test results can hide overfitting. The protocol is designed to make these shortcuts visible.

O.24 FRESH-AGENT NAVIGATION GUIDE

A new agent entering this repository should start from the following files:

- README.md: thesis, result directories, commands, and v4 artifact paths.

- docs/v4_execution_plan.md: strong-accept hardening plan.
- src/bayesian_ensemble_bon/selectors.py: selector families.
- src/bayesian_ensemble_bon/gym_benchmark.py: classic-control bridge.
- src/bayesian_ensemble_bon/mujoco_benchmark.py: MuJoCo bridge.
- experiments/run_v4_gym_benchmark.py and experiments/run_v4_mujoco_benchmark.py: final v4 regeneration scripts.
- scripts/build_paper.ps1: manuscript compilation and versioned final PDF export.

The source-of-truth final evidence lives in results/full, results/v3, results/v4, and results/v4_mujoco. Smoke directories are development evidence only. The paper macros are copied from the final directories during the build. A fresh agent should not infer final claims from old smoke runs or from the visible Desktop PDF alone.

O.25 FINAL SUBMISSION CHECKLIST

Before submission, the following checks should pass:

- all tests pass under `python -m pytest -q`;
- the paper builds from scratch using `scripts/build_paper.ps1`;
- the final PDF is versioned as v4 and copied to the visible Desktop;
- citation links are visible and unresolved citation warnings are gone after the final LaTeX passes;
- v4 protocol-freeze JSON files exist and contain artifact hashes;
- PAPER_SOURCE_MAP.md points the Desktop PDF to this source folder and GitHub repository;
- the repository is committed and pushed to GitHub after the PDF, source, results, and figures are synchronized.